

Quantum Gravitation

Problem Set 8

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Due 2026-10-25 18:00

Topic

Midterm Review

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Midterm Review. State all assumptions and conventions.

$$checksum = hash(C_{66}, grid, basis)$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$\Pi^{(2)} : \Pi^{(2)} = 5$$

3. Verify the third relation in two limiting regimes and explain the physical meaning of the result.

$$\Delta E_{num} < 0.01\sigma_{registry}$$

Show enough intermediate work that another student can reproduce the calculation.